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Pollen morphology in the genus *Coffea* (Rubiaceae).

II. Pollen polymorphism

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Pollen grains of 13 species and varieties investigated within *Coffea* L. (Rubiaceae) were found to be highly polymorphic. The majority are members of the Section *Eucoffea*, and two species belong to the Section *Paracoffea*. *Coffea arabica* L. and *C. eugenoides* Moore show the highest degree of polymorphism.

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While reflecting on the work of Müller (1869) on the genus *Faramea* (Rubiaceae), Darwin (1877) introduced the concept of variations in pollen size and morphology within similar plant species (vide Baker 1956). Over 100 years later, Müller's *Faramea* specimen has since been redetermined as *Rudgea jasminoides* (Baker 1956), and pollen dimorphism has been found to occur in several families. Rubiaceae is the most complex family, where pollen dimorphism is known from some 68 genera (Vuilleumier 1967).

During a survey of the pollen morphology of 38 representative species and varieties within the genus *Coffea* L. (Rubiaceae), 13 species and varieties have been found to be di-, tri-, tetra-, penta-, and polymorphic.

Broad or taxonomic surveys which include data on the pollen of *Coffea* are cited by Thanikaimoni (1972, 1973, 1976). Both Van Campo et al. (1964) and Chinnappa & Adams (1974) recognized a high degree of variation in the pollen of some species.

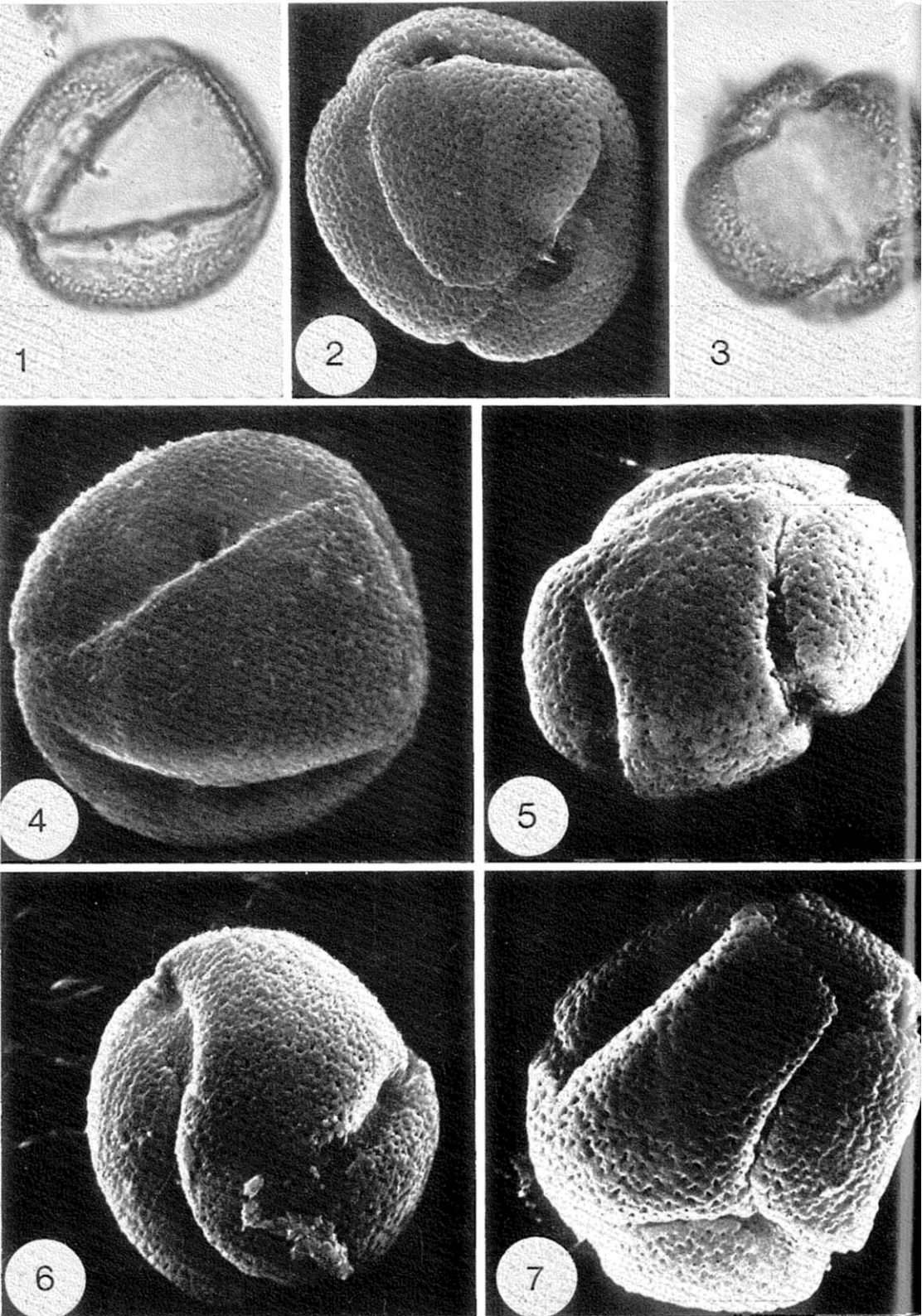
This paper reports and demonstrates the degree of pollen variability within some *Coffea* species through the use of light and scanning electron microscopy studies. Erdtman (1966) simply defines pollen polymorphism as plants having several kinds of palynomorphs. Polymorphic pollen grains may

occur on a single stamen, which suggests no genetically fixed stock. Alternatively, pollen polymorphism may occur in heterostylous plant species where the pollen of pin and thrum flowers differ, thereby suggesting two predetermined genetic stocks. Heterostyly is a common phenomenon in Rubiaceae (Vuilleumier 1967), and Burck (1884) had provided some evidence for heterostyly in *Coffea*.

The floral biology of *Coffea* is not well known; however, at present it seems likely that pollen polymorphism is a heteromorphic manifestation of an anti-self pollinating mechanism, represented by a continuum of aberrant palynomorphs. Our preliminary examination would lead us to believe that most *Coffea* species are probably not monomorphic *sensu stricto*, since the pollen of most species are either tri- and tetrazonocolporate, or tetra- and pentazonocolporate. At this time, we have chosen to focus on those species which have tri-, tetra-, or pentazonocolporate pollen grains with syncolpate forms and variations thereof.

MATERIALS AND METHODS

Pollen was obtained from specimens in the herbaria and from living plants at the U.S. National Arboretum, Washington; Coffee Rust Research Centre, Portugal; Uni-



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versity of Waterloo, Canada, and Central Coffee Research Institute, India.

Anthers were subjected to acetolysis treatment (Fægri & Iversen 1975) and subsequently split in half for LM and SEM analyses. Pollen for LM was dehydrated in an ethyl alcohol series, tertiary butyl alcohol and mounted in silicone oil. SEM preparations followed Adams & Morton (1972).

Light microscope observations were made in bright field on a Carl Zeiss microscope at $\times 400$ magnification and under oil immersion at $\times 1000$ magnification. The polar axis (P) and equatorial diameter (E) were measured on the most frequent zonocolporate pollen type of each taxon. Between 10 and 20 measurements were taken along each axis. P/E ratios were calculated according to Erdtman (1969).

Frequency of occurrence of each pollen type within each species was estimated from a minimum total of 100 grains encountered while scanning over the slide. The SEM studies were employed to provide the details of morphology beyond the resolution of LM.

Descriptive terminology follows Erdtman (1969).

RESULTS

Section Eucoffea

Coffea arabica L.

C. arabica has the most variable pollen of all *Coffea* species we have studied. Six distinct pollen types and variations have been encountered (Table I).

A. Nomotreme forms:

Zonocolporate grains are isopolar, spheroidal, radially symmetrical, tri- or tetrazonocolporate. Amb is circular to semi-angular. P/E is 0.97, P: $33.8\ \mu\text{m}$ (31–38 μm) and E: $32.9\ \mu\text{m}$ (30–36 μm). Prominent colpi have irregular margins which gradually taper meridionally. Tectum merges with the colpus membrane. Each os is round to irregular. The exine is tectate, psilate-scabrate with scattered puncta, perforate to foveolate to finely reticulate with suprategal processes.

Figs. 1–7. C. arabica, pollen type A showing pollen grain in side view with diverging meridional compound colpi ($\times 1300$). 2. *C. arabica*, pollen type A. Short diverging meridional compound colpi joined with simple colpi ($\times 2500$). 3. *C. arabica*, pollen type A. Four compound colpi anastomose with single, simple colpi (out of focus) ($\times 1300$). 4. *C. arabica*, pollen type with small, reduced ora ($\times 2500$). 5. *C. arabica*, pollen type B showing compound colpi anastomosing with two simple colpi ($\times 2500$). 6. *C. arabica*, pollen type C with two opposing $\pm$ spiral apertures. Leptoma in upper endoaperture of photograph and ora in other endoaperture ($\times 2500$). 7. *C. arabica*, pollen type D showing irregular form and arrangement of endoapertures. ($\times 2500$).

Table I. Frequency of pollen types in *C. arabica* (N=100)

	Per cent
Nomotreme forms:	
Trizonocolporate	43.0
Tetrazonocolporate	5.3
Anomotreme forms:	
Type A	49.5
Type B	5.3
Type C	<1.0
Spirotreme form:	
Type D	<1.0

Occasionally grains are polarly syncolpate in trizonocolporate forms. In tetrazonocolporate grains, there is either a simple colpus or a fold in the sexine. This is fused to the ends of adjacent compound colpi. These "simple colpi" are not regularly positioned, and the number may vary from 3 to 8 per grain. This pollen type occurs quite frequently in other *Coffea* species as well.

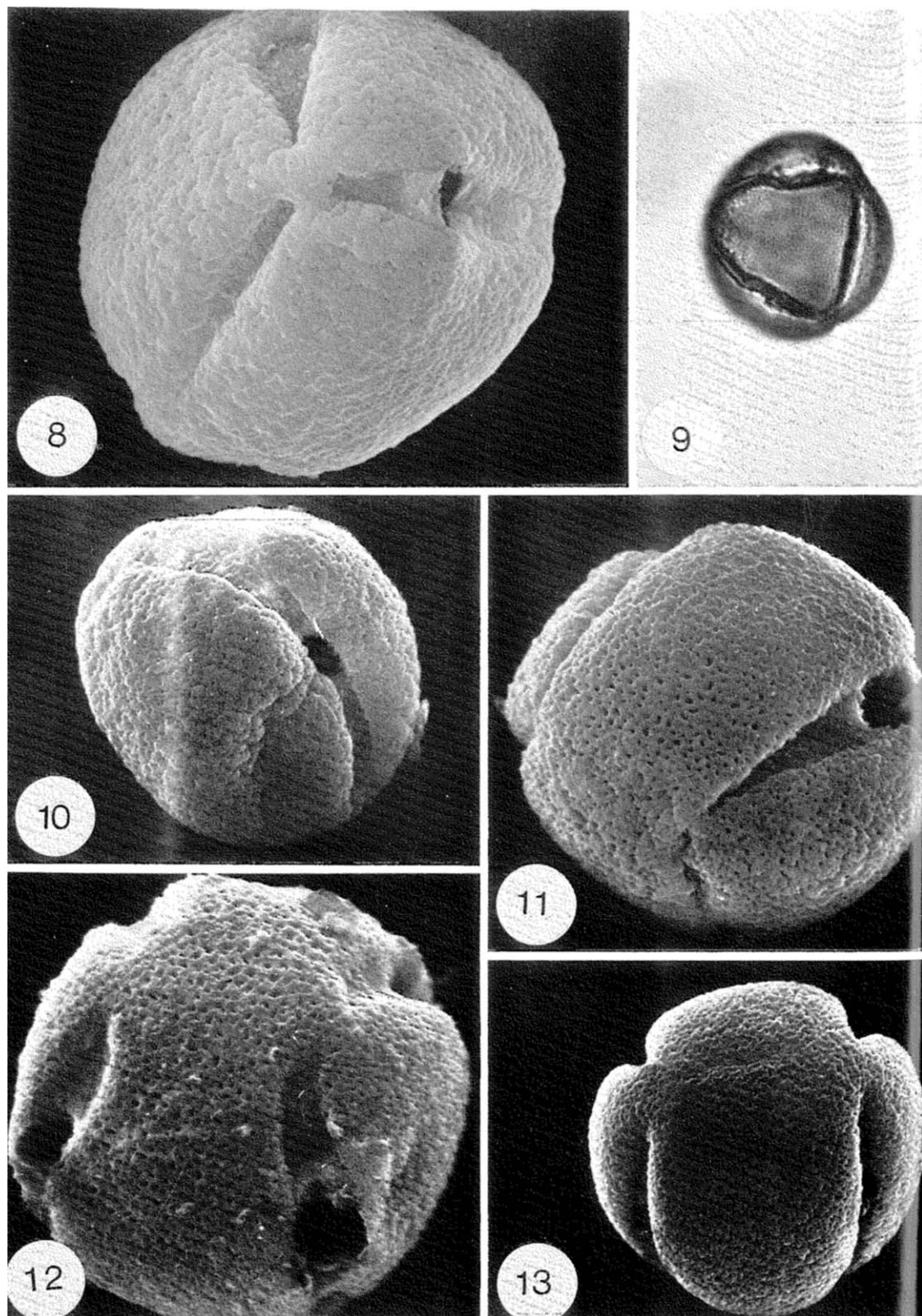
B. Anomotreme forms:

Type A (Figs. 1–4): Grains are radially symmetrical, heteropolar and vary in shape from spheroidal to slightly spheroidal, subprolate. Four meridional colpi may or may not converge with a simple, narrow, slit-like colpus which runs across the poles (Fig. 2). Colpus margins are more or less parallel along their entire length. An os is midway on each of the meridional colpi, and may be obscure or poorly developed. Ora are circular to elongate and variable in size, even on the same grain (Fig. 4). Exine is tectate, perforate to foveolate.

Type B (Fig. 5): This pollen type is radially symmetrical, heteropolar, tetrapantocolporate, and spheroidal to spheroidal, suboblate.

Occasionally adjacent compound colpi are joined by a slit-like simple colpus. The number and position of these on the grain are variable. There is a small, round costate os positioned equatorially on each colpus. Exine is tectate, and densely punctate to reticulate.

Type C (Fig. 6): Grains are spheroidal and polypantocolporate. The exact form of this grain is difficult to define; it may actually be "pentapantocolporate" where simple colpi delineate the apocolpium from 5 meridional compound colpi. Colpus margins are more or less parallel and of different lengths. Irregular shaped ora are



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positioned at various points along the colpi. The number of ora per grain is variable. Exine is tectate and foveolate.

C. Spirotreme forms:

Type D (Fig. 7): Grains are spheroidal, $\pm$ spiraperturate, with two opposing and spirally positioned endoapertures. Aperture margins are more or less regular. Tectum gradually merges with the nexine. Four small ora are positioned on a single plane midway on the pollen face. Occasionally ora appear as leptoma in the colpus membrane. Exine is tectate, perforate to foveolate.

Coffea congestis Frohener

Grains are isopolar, spheroidal, tri- (89%), or tetrazonocolporate (1.8%), and radially symmetrical. Amb is circular. P/E is 0.88, P is 24.3 μm (23–26 μm) and E: 27.4 μm (23–27 μm). Colpus margins are more or less regular. Tectum gradually merges with the colpus membrane. Colpi taper meridionally. Large costate round ora are slightly lolongate. Exine is tectate, foveolate to reticulate with supratectal processes. Polarly syncolpate forms are present in both tri- (3.6%) and tetrazonocolporate (1.8%) types.

Radially symmetrical, heteropolar types occurred 2.7% (Fig. 9). Three compound meridional colpi nearly merge around a small apocolpium at one pole, and are fused with 3 narrow, simple colpi defining a large apocolpium at the opposite pole.

Coffea eugenoides Moore

Grains are isopolar, spheroidal, radially symmetrical, tri- and tetrazonocolporate (36.8% and 2.6% respectively). Amb is circular to subtriangular. P/E is 0.99, P 26.4 μm (25–29 μm) and E 26.3 μm (22–29 μm). Distinct and regular colpi taper to a small apocolpium (Fig. 8). Round, costate ora vary in

size. Colpus margin may be constricted and overlaps a small os in some grains. Exine is tectate, finely reticulate, to foveolate with scattered supratectal processes.

Polarly syncolpate grains (1.8%) may or may not have a small triangular apocolpium (as in plate 116, nos. 5–8 in Van Campo et al. 1964).

The most frequent pollen type (55.26%) is hexapantocolporate-syncolpate grains. The grains may be syncolpate in being joined at any 1–4 points, and not all 3 colpi are necessarily fused at each point. The number of ora may vary from 3 to 6 per grain, and if present, the os is midway on each colpus.

Rarely, there are tetrapantocolporate grains (Fig. 11). Colpi are broad, tapering and anastomosing at one end of the grain. The sexine is broken at the opposite end by a large, wide linear endoaperture. A similar pollen form was observed by Chinnappa and Adams (1974: fig. 1 c) in *C. arabica*.

Coffea ligustroides S. Moore

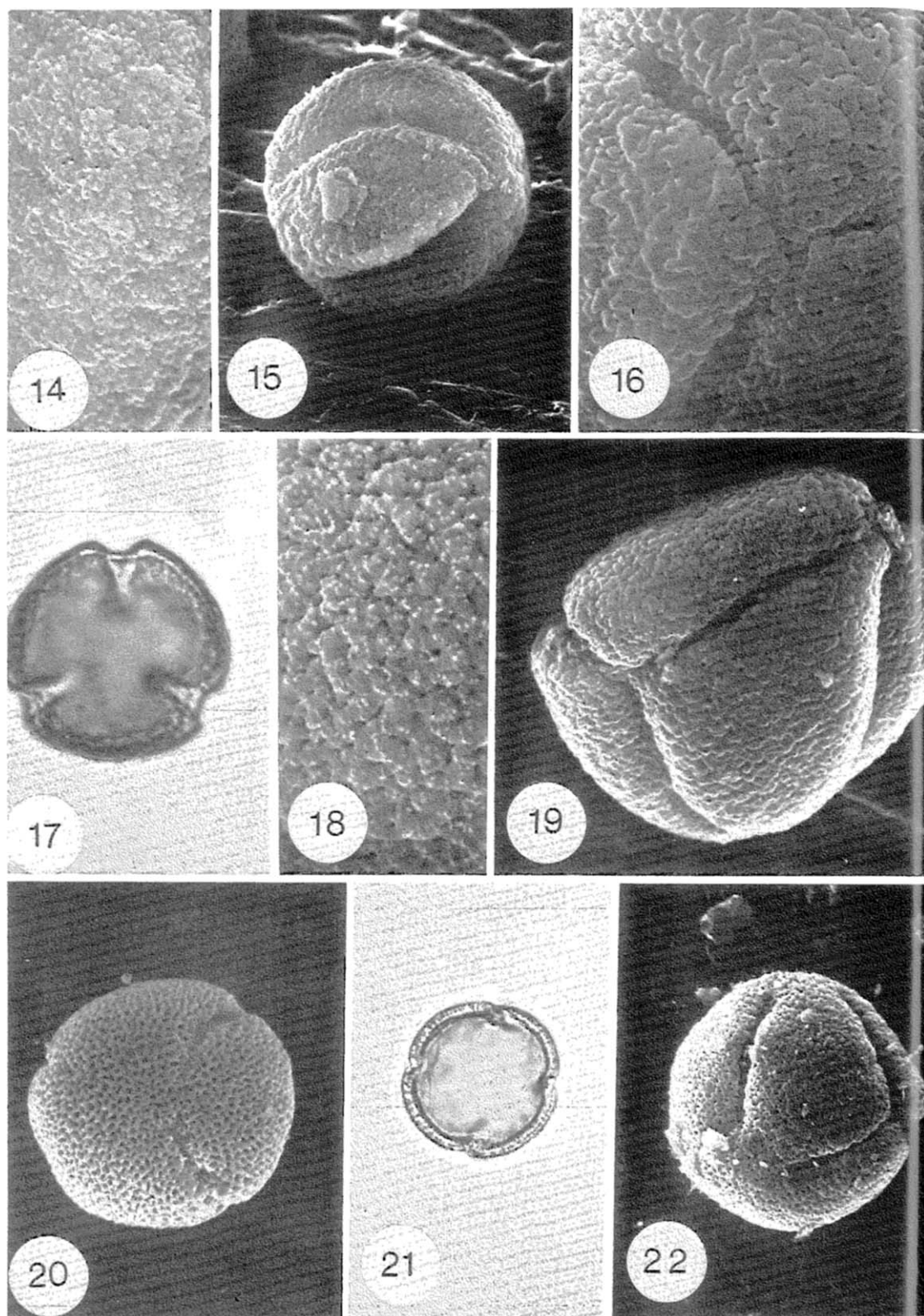
The pollen is primarily trizonocolporate (78.4%), isopolar, spheroidal to spheroidal subprolate in shape, and radially symmetrical. Amb is circular. P/E is 1.15, P 28.6 μm and E 24.7 μm (23–27 μm). Colpi margins are well defined, with blunt to tapering ends. Apocolpium is small. Ora are costate, round to lolongate (Fig. 10). Exine is tectate, foveolate to reticulate with supratectal processes. Polarly syncolpate forms occur occasionally.

About 21% of the grains appear to be syncolpate (as in *C. arabica*, pollen type A). Four compound transverse colpi and anastomose at the poles by a simple colpi.

Coffea salvatrix Swynn. & Phil.

Most of the pollen (92.8%) is trizonocolporate. Less than 1% are tetrazonocolporate and 6.2% are syncolpate (as in *C. arabica*, pollen type A). Grains are isopolar, spheroidal to spheroidal subprolate and radially symmetrical. Amb is circular. P/E is 1.08, P is 25.1 μm (22–27 μm), and E: 23.1 μm (22–25 μm). Colpi are wide with regular margins. In trizonocolporate grains; colpi gradually taper meridionally to a small polar area, and in tetrazonocolporate grains the colpi are wide, short and blunt with a large apocolpium (Fig. 12). Colpi margins are constricted at each os. Ora are costate, and small in trizonocolporate grains, and much larger in

Figs. 8–13. 8. *C. eugenoides*, trizonocolporate grain with small apocolpium ($\times 3000$). 9. *C. congestis*. Side view of heteropolar syncolpate grain with two transverse compound colpi, converging to a small apocolpium at one pole, and diverging and fused with simple colpi at a large apocolpium ($\times 1300$). 10. *C. ligustroides*. Note round, costate lolongate ora ($\times 2500$). 11. *C. eugenoides*. Tetrapantocolporate pollen showing broad, tapering anastomose compound colpi and no joining fold or simple colpi ($\times 2500$). 12. *C. salvatrix*. Tetrazonocolporate pollen form with broad colpi and large costate ora ($\times 2500$). 13. *C. salvatrix*. Syncolpate grain with no anastomosing fold ($\times 2500$).



tetrazonocolporate grains. Exine is tectate and finely reticulate.

Syncolpate grains resemble *C. arabica*, pollen type A. Four transverse compound colpi meet but do not fuse with a simple colpus or fold in the exine. Occasionally this colpus or fold is poorly developed, and even absent in some cases (Fig. 13). Ora are costate, round and much reduced.

Coffea brevipes Hiern

Isopolar, spheroidal, spheroidal suboblate, radially symmetrical, tri- (64.6%) and tetrazonocolporate (15.1%). Amb is circular, to slightly triangular. P/E is 0.92, P 25.3 μm (23–27 μm) and E 27.4 μm (24–31 μm). Colpi are well-defined, narrow and taper meridionally to a small apocolpium. Ora are round to irregular. Exine is tectate, perforate with supratectal processes (Fig. 14).

Syncolpate forms (20%) are similar to *C. arabica*, pollen type A. Four meridional compound colpi are joined by a simple narrow colpus at each pole.

Coffea humilis Chev.

Grains are isopolar, radially symmetrical, spheroidal to spheroidal subprolate, tri- (64.2%) and tetrazonocolporate (3.06%). Amb is circular. P/E is 1.01, P 20.4 μm (19–23 μm) and E 20.1 μm (17–23 μm). Colpi are irregular, narrow, and tapering to a small apocolpium. Exine gradually merging with the colpus membrane. Ora are small, round irregular, and occasionally rather obscure. Exine is tectate and perforate with supratectal processes.

Syncolpate forms (32.6%) have 4 compound colpi (as *C. arabica* pollen type A) which may or may not converge, but do not fuse with a fold, or simple slit-like colpus at either one or both ends of the grain (Fig. 15).

Figs. 14–22. 14. *C. brevipes*. Perforate sculpturing with supratectal processes ($\times 7000$). 15. *C. humilis*. Syncolpate pollen ($\times 2500$). 16. *C. abeokutae*. Three anastomose colpi in polarly syncolpate pollen form of trizonocolporate grains ($\times 7000$). 17. *C. dewevrei*. Three broad colpi tapering to a small apocolpium in trizonocolporate grains ($\times 1300$). 18. *C. dewevrei*. Foveolate sculpturing with supratectal processes ($\times 7000$). 19. *C. dewevrei* var. *excelsa*. Tripantocolporate pollen form ($\times 2500$). 20. *C. travancorensis*. Two polarly anastomose colpi in trizonocolporate pollen ($\times 2500$). 21. *C. travancorensis*. Tetrazonocolporate pollen showing tectate exine and prominent columella ($\times 1300$). 22. *C. wightiana*. Hexapantocolporate pollen form showing irregularity in anastomosing colpi ($\times 2500$).

Coffea abeokutae Cramer

In this species 66.4% have trizonocolporate pollen and 12.6% are polarly syncolpate forms of these; 10.4% tetrazonocolporate and 11.2% were polarly syncolpate.

Isopolar, spheroidal, radially symmetrical, tri- and tetrazonocolporate. Amb is circular. P/E is 1.0, P 20.0 μm (18–22 μm) and E 19.8 μm (18–22 μm). Colpi are narrow, and taper meridionally to a small apocolpium. Colpus margins are irregular with the exine gradually merging with the colpus membrane, particularly polarly where the colpus margin is ill-defined. Ora are costate and irregular. Occasionally ora protrude slightly. Exine is tectate, perforate to foveolate. Polarly syncolpate forms are present in both tri- and tetrazonocolporate grains (Fig. 16).

Coffea dewevrei De Wild. & Durand

The pollen of *C. dewevrei* was studied by Moens (1962) who encountered only tricolporate pollen. Trizonocolporate pollen occurred 82%, and 17.7% were tetrapantocolporate.

Isopolar, spheroidal to spheroidal subprolate, radially symmetrical, and trizonocolporate. Amb is circular to subtriangular. P/E is 1.15, P: 25.3 μm (24–27 μm) and E: 22.1 μm (22–25 μm). Colpi are well-defined, broad and taper meridionally to a small apocolpium (Fig. 17). Ora are costate, and circular to lolongate. Exine is tectate, foveolate with supratectal processes (Fig. 18).

C. dewevrei var. *aruwimensis* (De Wild.) Chev.

Trizonocolporate grains were the most common types (91.3%). Grains are isopolar, spheroidal, spheroidal suboblate radially symmetrical. Amb is circular. P/E is 0.96, P 22.6 μm (21–24 μm) and E 23.6 μm (21–24 μm). Colpi, ora and sculpturing are as in *C. dewevrei*. There are rarely polarly syncolpate forms. Tetrapantocolporate forms occurred 5.7% (as in *C. dewevrei*) and 1.9% were hexapantocolporate.

C. dewevrei var. *excelsa* Chev.

Grains are isopolar, spheroidal, spheroidal suboblate, radially symmetrical, and trizonocolporate (32.7%). Amb is circular. P/E is 0.93, P 22.0 μm (19–24 μm) and E 23.7 μm (21–26 μm). Colpi are narrow, regular, and tapering meridionally. Ora are small, round to slightly lalongate, regular or irregular, and sometimes obscure. Exine is tectate, punctate to foveolate with supratectal processes.

The most common pollen type (50%) is a radially symmetrical, heteropolar (Fig. 19), tripantocolporate form (as in the *C. congensis* syncolpate type). Three compound colpi anastomose with 3 folds or narrow, simple colpi at one pole forming a large apocolpium. Tetrapantocolporate (17.3%) forms with 4 compound colpi also anastomose with 4 folds, or narrow simple colpi at one pole. These compound colpi taper to a small apocolpium at the opposite pole.

Section Paracoffea

Coffea travancorensis Wight & Arn.

Grains are isopolar, spheroidal to spheroidal subprolate, radially symmetrical, primarily trizonocolporate (17%) and tetrazonocolporate (67%) (Fig. 21). Amb is circular. P/E is 1.03, P 19.9 μm (18–21 μm) and E 19.2 μm (17–20 μm). Colpi are short, narrow and poorly defined, often appearing as a slit in the exine. Ora are very small and obscure, round to irregular. Exine is tectate and reticulate with supratectal processes.

Polarly syncolpate forms were present in tri- (3%) and tetrazonocolporate (12%) grains. Rarely two adjacent colpi in trizonocolporate grains are fused at the poles (Fig. 20).

Coffea wightiana Wall.

Grains are isopolar, spheroidal to spheroidal subprolate, radially symmetrical, tetrazonocolporate (86%). Amb is circular. P/E is 1.06, P 20.0 μm (18–22 μm) and E 18.8 μm (18–19 μm). Colpi are short, narrow, and irregular with a large apocolpium. Ora are small, round to elongate, and slightly irregular. Exine is tectate and reticulate with scattered supratectal processes. Tetrazonocolporate grains may be polarly syncolpate (6.7%) with short and narrow slit-like simple colpi.

Hexapantocolporate forms (11.4%) are similar to *C. arabica*, pollen type A, in the arrangement of the colpi. There is some variability in the number and position of compound apertures, and in the number of colpi fused at any one point (Fig. 22).

DISCUSSION

The genus *Coffea* is classified into four sections consisting of approximately 60 species. Among 38 *Coffea* species investigated, 11 are distinctly polymorphic of which 9 belong to sect. *Eucoffea*.

The remaining two species belong to sect. *Paracoffea*.

Moens (1962) described the pollen of *C. congensis* and recognized tricolporate, and rarely tetracolporate types. Van Campo et al. (1964) recognized tri- and tetracolporate grains, as well as polarly syncolpate grains, with a triangular apocolpium in tricolporate grains to be very frequent. We found not only tri-, and tetrazonocolporate grains, but also various syncolpate forms. The pollen of *C. eugenioides* is very plastic in both form and size. Our observations agree with these previous observations, but in addition there are tetra-, and hexapantocolporate grains which may or may not be syncolpate. Only tricolporate grains were observed in *C. abeokutae* by Moens (1962). We observed tri-, and tetrazonocolporate grains, and polarly syncolpate forms of these in this species. The pollen of *C. arabica*, and to a lesser degree, of *C. eugenioides* show extreme pollen polymorphism, with some pollen types in the two species being quite similar.

The implications of these findings are quite interesting both in terms of the pollination biology and the taxonomy of the genus; the latter will be discussed in a forthcoming paper. Our studies have encompassed only about half of the known *Coffea* species. As palynological studies are expanded to other species of *Coffea*, new pollen types and the number of *Coffea* species with pollen polymorphism may emerge.

Pollen morphology within *C. arabica* and its 48 known varieties and cultivars, alone remains a challenging study which is, as yet virtually unexplored. The heterogeneity of pollen forms, and particularly unusual types such as the spiraperturate and polypantocolporate pollen in *C. arabica* reflects on its genetic constitution. The origin of this commercially important tetraploid species is still open to speculation.

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